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(Autonomous Institute, Affiliated to VTU)
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SEMESTER END EXAMINATIONS FEBRUARY - MARCH 2021

Program	: B.E.: Information Science and Engineering	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: IS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT- I

- Briefly discuss the four basic functions and Structural components that enables the performance of a computer. CO1 (08)
 - Briefly discuss on Multicore, MICS and GPGPUS with its usage in the computer system. CO1 (06)
 - Suppose that a task makes extensive use of floating-point operations, with 40% of the time consumed by floating-point operations. With a new hardware design, the floating-point module is sped up by a factor of K. Find the overall speedup of the system. CO1 (06)
- Briefly discuss the various techniques that are built into a contemporary processor by the designers to fabricate chips of greater density. CO1 (06)
 - Discuss the Amdahl's law that provides an insight into the performance of parallel processing systems. CO1 (06)
 - Briefly discuss the Bus Interconnection scheme with appropriate diagram with all the components involved in the design process. CO1 (08)

UNIT – II

- Discuss the method of accessing units of data among memory along with the three performance parameters of a memory. CO2 (06)
 - Briefly discuss the three level cache organization in a memory structure. CO2 (06)
 - Briefly discuss the data organization and formatting on a disk that are capable on reading from or writing to a portion of the platter. CO2 (08)
- Briefly discuss the memory hierarchy with appropriate diagram that are required for better performance of a system. CO2 (07)
 - Discuss the various categories of major functions or requirements for an I/O module. CO2 (07)
 - Write short notes on Direct Memory Access, mentioning its importance in the computer system. CO2 (06)

UNIT – III

- Using Quine-McClusky Technique, minimize the Boolean Function $F(A,B,C,D)=\sum(0,2,4,5,8,9,12)$. CO3 (10)

- b) Design a sequential circuit to the sequence 000011010010001000 using positive edge triggered IS flip-flop whose function table is given below CO3 (10)

I	S	$Q_{(t+1)}$
0	0	1
0	1	Q
1	0	Q'
1	1	0

6. a) Design a synchronous counter using JK flip-flops by determining the values of inputs J and K by using the excitation table. CO3 (10)
- b) Describe using a flowchart the concept of the Booth's algorithm for two's complement multiplication. Given $x=0101$ and $y=1010$ in two's complement notation compute the product $p = x * y$ with Booth's algorithm. CO3 (10)

UNIT – IV

7. a) What are the typical elements of a machine instruction? Describe the types of locations can hold source and destination operands. CO4 (06)
- b) Explain the advantages and limitations of each addressing modes. CO4 (07)
- c) Describe the effect of a conditional branch on instruction pipeline operation with a neat timing diagram. What do you mean by branch penalty? CO4 (07)
8. a) Explain different types of operations and operands with examples. CO4 (10)
- b) List and explain different types of registers. CO4 (05)
- c) With a neat diagram, describe the simplified view of a processor, indicating its connection to the rest of the system via the system bus. CO4 (05)

UNIT – V

9. a) List the important characteristics of SMP. Explain the advantages of SMP over uniprocessor. CO5 (08)
- b) Define GPU. Compare CPU and GPU with example. CO5 (06)
- c) Explain different micro-operations. CO5 (06)
10. a) Explain the four states in MESI protocol with neat diagram. CO5 (10)
- b) Describe different methods of clustering with benefits and limitations. CO5 (10)
